



PROGRAMMING FUNDAMENTALS

CS – 101

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CONDITIONS

- Execution occurs on the basis of some specified conditions
- Conditions are logical comparisons between two values
- If the condition is met, the instruction(s) following it are executed
- Otherwise, the instruction(s) are skipped
- Logical operations are used to check for conditions

MODULUS OPERATOR

- Works on integers
- Yields remainder when first operand is divided by the second
- Python uses % sign to perform the operation
- Examples:
 - `quotient = 7 // 3`
 - `print(quotient)` → Output = 2
 - `remainder = 7 % 3`
 - `print(remainder)` → Output = 1
 - What does it mean if `x % y = 0`? → x is completely divisible by y
 - What will `x % 10` achieve? →

COMPARISON OPERATORS

- Relational operators are used to check for arithmetic comparison
 - == → Equals to
 - != → Not equals to
 - < → Less than
 - <= → Less than or equal to
 - > → Greater than
 - >= → Greater than or equal to

BOOLEAN EXPRESSION

- An expression that is either ***True*** or ***False***
- ***True*** and ***False*** are special values that belong to type ***bool***
- They are not string or text values, but a category
- Examples:
 - `x = 5`
 - `x == 6?` → output = ?
 - `x == 5?` → output = ?
 - `type(True)` → output = ?
 - `type("True")` → output = ?
 - `type(False)` → output = ?
 - `type("False")` → output = ?

LOGICAL OPERATORS

- ***and***

- Needs all values to be **true** for the condition to work
- Example:
 - if `(x > 0) and (x < 10)`:
 - This will only work if both conditions are satisfied, that is, x is greater than 0 and x is also less than 10

- ***or***

- Needs either one of the conditions to be **true** for it to work
- Example:
 - if `(n % 2 == 0) or (n % 3 == 0)`:
 - This will work only if either n is divisible by 2 or 3

- ***not***

- Checks for negated Boolean condition
- Example:
 - if **not** `(x > y)`:
 - This will work if x is NOT greater than y

CONDITIONAL EXECUTION

- In order to write useful programs, it must be able to check conditions and execute accordingly
- ***Conditional statements*** provides that capability

Boolean expression

```
if (x > 0):
```

conditional statement

Body { **print ("x is positive")**

- If there are cases where you don't want the program to do anything, you can use the keyword ***pass***

ALTERNATIVE EXECUTION

- When there are more than one possibilities for conditions
- Keyword to check for the alternate condition: ***else***
- Example:
 - ```
if (x % 2 == 0):
 print ("x is even")
else:
 print("x is odd")
```
- Used only when there are exactly two choices
- The alternatives are called **branches**



# CHAINED CONDITIONALS

- More than two branches required
- Keyword to check for those condition: *elif*
- Example:
  - ```
if (x < y):  
    print("x is less than y")  
elif (x > y):  
    print("x is greater than y")  
else:  
    print("x and y are equal")
```
- Not always necessary to have else in the end

NESTED CONDITIONALS

- One condition nested within another
- There's **outer condition** and then there's **inner condition(s)**
- Example:
 - ```
if (x == y):
 print("x and y are equal")
else:
 if (x < y):
 print("x is less than y")
 else:
 print("x is greater than y")
```

# NESTED CONDITIONALS

- Both inner and outer conditions can have multiple branches
- Use correct indentation to define the outer and inner conditional blocks clearly
- Avoid using nested conditions
- Example:
  - ```
if (0 < x):  
    if (x < 10):  
        print("x is positive")
```
 - The above example can be written more efficiently as:

```
if (0 < x) and (x < 10):  
    print("x is positive")
```

SELF-PRACTICE TASK

- Hackerrank Link: <https://hr.gs/cs102Fall2022SPTaskCh5Conditions>
- These questions will be available until next Friday, September 16th, 2022, so do work on them, as well as all the other practice problems being shared by the other faculty and staff.
- These are Self Practice Tasks so they will not be graded. If you have any “questions about the questions” 😊 you can come see me during my office hours:
 - Tuesdays/Thursdays: 2:00pm – 3:00pm
 - Office: W-230